

RESEARCH NOTE

AMDR-Li: what the results actually mean

A plain-English one-page summary of the core-paper replication and the lithium extension

Bottom line. AMDR-Li is the strongest model on RMSE, meaning it reduces the largest forecasting misses. It is not yet a proven real-time trading system or a universal winner on every metric.

1. Why an 80:20 holdout is not real-world implementation

Imagine 100 historical trading days. A static 80:20 study builds the model using the first 80 days, then reports results on the final 20. That is useful as a laboratory comparison, but a live forecaster must make every decision using only information known before each next-day forecast.

What both studies lack: The core paper and our current AMDR-Li paper share this limitation. Full-sample decomposition, global preparation and observed-test ARIMA filtering use later observations. That makes the headline results retrospective evidence, not a simulation of decisions made day by day in real time.

2. What happened when we replicated the fixed older method

ARIMA is a statistical model that projects the next value from a component's recent history; it is useful for smoother, more linear patterns. LSTM is a neural-network sequence model; it can learn more complex nonlinear patterns but needs more data and tuning. The fixed older recipe always sends the first low-frequency components to ARIMA, all remaining components to LSTM, then adds the forecasts. On lithium, that rigid rule did not produce a universal win.

3. What AMDR-Li changes

The actual difference: AMDR-Li keeps the same broad decomposition idea, but replaces the core paper's fixed routing rule. It tests VMD and CEEMDAN candidates, selects features separately for each component with LASSO, then lets development-period error choose ARIMA, LSTM, SVR or Random Forest for each component. It combines the strongest candidate forecasts with weights fixed before the final 20% is scored.

Multimodal means: it uses lagged prices plus linked lithium prices, producer equities, lithium-specific search interest and news sentiment, sector indices and market controls. Li simply identifies the lithium application.

4. How good is it - and what still needs work

Target	RMSE result	Where it is weaker	Trading / publication reading
Carbonate	Best RMSE 2.881	MAE, MASE, MAPE, sMAPE and direction trail the best comparators.	AMDR-Li Scheme 1: Sharpe 7.81, but VMD-ARIMA is stronger (11.52).
Hydroxide	Best RMSE 0.274	Direction is only tied-best (44.29%); Random Walk has lower MAE and MAPE.	Sharpe 4.56; CEEMDAN-ARIMA is marginally higher (4.68).
ALB equity	Best RMSE 2.328	No clear point-metric loss within the tested set.	Strong extension; 53.67% lower RMSE than Random Walk.
Ganfeng	Best RMSE 0.190	VMD-ARIMA has better MAE, MASE, MAPE and sMAPE.	Useful but narrow RMSE gain: 0.55% over VMD-ARIMA.

Trading caution: the Sharpe ratios, returns and drawdowns are descriptive only. Because the study is retrospective, they cannot yet be treated as investable performance or evidence of abnormal return.

Publication judgement: This can support a transparent retrospective methodology paper, but it is not yet sufficient for a live forecasting or trading claim. We did run a past-only walk-forward test on the earlier Glencore critical-minerals pipeline: its hybrid RMSE was 0.240 versus 0.112 for Random Walk, so the advantage did not survive. Current AMDR-Li lithium targets still need that equivalent causal test, plus benchmark-relative return and factor-adjusted risk evidence.